



Ministry of
Education
Government of India



MoE's
INNOVATION CELL
(GOVERNMENT OF INDIA)



SMART INDIA
HACKATHON
2026

SMART INDIA HACKATHON · 2026

Easy **vs** Difficult Problem Statements

5 of the most beginner-friendly and 5 of the most technically demanding problem statements, pulled straight from the official list of 226 SIH 2026 PS — so you can pick the right level for your team.

How these were picked: the SIH portal doesn't publish an official difficulty rating, so every one of the 226 PS was scored on hardware requirement, technical keywords (AI/ML, sensors, real-time, robotics, cryptography etc.) and description depth. The lowest-scoring and highest-scoring statements below are shown as-is from the official listing.

226

TOTAL PS REVIEWED

172+54

SOFTWARE + HARDWARE

5+5

EASY + DIFFICULT PICKS

Easiest 5 — good for first-time teams

Pure software, CRUD-style portals with a clear, bounded scope. No AI research, sensors or hardware needed to reach a working demo.

Hardest 5 — for advanced / veteran teams

Defense hardware, robotics, real-time perception and adversarial ML — genuinely research-adjacent problems from DRDO, Qualcomm and MathWorks.

BEGINNER-FRIENDLY

The 5 easiest problem statements

Pure software builds with a clear, bounded scope — each one is a well-known software pattern (portal, LMS, compliance checker) that a first-time SIH team can realistically finish.

01 Portal for Academia–Industry Collaboration (Skill Mapping, Internships & Placement)

SIH26044 · Ministry of Ayush · Miscellaneous

Connects students, industries and academicians on one platform so skill gaps, internships and placements stop being guesswork.

SOFTWARE

EASY

02 Packaged Commodities Compliance Checker (Legal Metrology Rules, 2011)

SIH26034 · Ministry of Consumer Affairs, Food & Public Distribution · Agri, FoodTech & Rural Dev.

Scans product labels/images and checks them against mandatory declarations — MRP, net quantity, manufacturer details — automatically.

SOFTWARE

EASY

03 Online Verification System for Weighing & Measuring Instruments

SIH26036 · Ministry of Consumer Affairs, Food & Public Distribution · Transportation & Logistics

Digitises the paper-based process of applying for, scheduling and certifying verification of shop scales and measuring tools.

SOFTWARE

EASY

04 Integrated Polar Science Outreach, Knowledge Repository & Media Dissemination Portal

SIH26063 · Ministry of Earth Sciences (NCPOR) · Space Technology

A searchable archive for polar expedition reports, datasets, photos and videos, with auto-generated content for social media.

SOFTWARE

EASY

05 CAPACITY CONNECT — Digital Capacity Building & Learning Management Portal

SIH26075 · Ministry of Earth Sciences · Smart Education

A classic 3-role LMS (Trainee / Trainer / Admin) for courses, quizzes, recorded lectures and progress tracking.

SOFTWARE

EASY

Why these are easy: every one of them is a bounded CRUD/portal build — logins, listings, forms, search, and a certificate/report generator. No model training, no hardware, no real-time pipeline stands between you and a working demo.

EASY · DEEP DIVE

Problem → Why Easy → Approach → Tech Stack → MVP

Everything needed to draft the idea PPT for each of the 5 easiest picks.

01 Academia–Industry Portal SIH26044 · Ministry of Ayush PROBLEM Students, industries and academicians all lack visibility into each other's skills, openings and internship capacity. WHY EASY Three-role signup + listings + search — no AI or hardware needed for a working version. APPROACH Role-based portal with skill-tagged profiles, an internship board, and an application tracker. TECH STACK React/Next.js, Node/Django, PostgreSQL, keyword-based matching. MVP Working signup for all 3 roles + one searchable internship board.	02 Compliance Checker SIH26034 · Ministry of Consumer Affairs PROBLEM Every packaged product must legally show MRP, net quantity and manufacturer info; manual checking doesn't scale. WHY EASY A bounded "scan → OCR → checklist match" loop, not an open-ended AI system. APPROACH OCR extracts label text, a rule engine checks it against the mandatory-declaration checklist. TECH STACK OCR API (Tesseract/Cloud Vision), Python rule engine, simple capture app. MVP Upload one product photo → get a pass/fail against 5 mandatory fields.
03 Instrument Verification System SIH26036 · Ministry of Consumer Affairs PROBLEM Verifying weighing/measuring instruments is a fully manual, paper-based application-to-certificate process today. WHY EASY Classic e-governance workflow digitisation — forms, scheduling, certificates — no novel tech. APPROACH Apply → auto-scheduled slot → digital certificate → automated renewal reminders. TECH STACK Standard web stack, PDF certificate generation, email/SMS reminders. MVP One flow: apply → get a slot → download a digital certificate.	04 Polar Science Outreach Portal SIH26063 · Ministry of Earth Sciences (NCPOR) PROBLEM NCPOR's expedition reports, datasets and media are scattered, making outreach and knowledge-sharing hard. WHY EASY A content/media repository with search and tagging — no modeling or real-time systems involved. APPROACH Searchable, categorised archive with an admin CMS and auto-generated social snippets. TECH STACK Headless CMS (Strapi/WordPress), Elasticsearch/Algolia search, static frontend. MVP A searchable archive for one expedition's reports, photos and datasets.
05 CAPACITY CONNECT (LMS Portal) SIH26075 · Ministry of Earth Sciences PROBLEM Organizational training happens ad hoc, with no central system to run courses, track progress, or share resources across trainees, trainers and admins. WHY EASY A textbook Learning Management System — three roles, courses, quizzes, tracking — a well-trodden software pattern with a clear scope. APPROACH Trainee/Trainer/Admin roles, course creation, MCQ assessments, recorded-lecture uploads, and a progress dashboard. TECH STACK Django/Rails or MERN stack, role-based auth, S3 + video player for lectures, a simple quiz engine. MVP One course with a lecture upload, one MCQ quiz, and a trainee progress view.	

ADVANCED TEAMS ONLY

The 5 hardest problem statements

Defense hardware, live robotics, adversarial ML and underwater perception — posted by DRDO, Qualcomm and MathWorks. Pick these only if your team has shipped hardware or ML systems before.

01 High-Altitude Performance Optimization of Anti-Drone Systems

SIH26050 · DRDO · Defense / Hardware

Anti-drone systems that work at sea level fail at high altitude — extreme cold, thin air and wind degrade mechanical, RF and optical subsystems.

HARDWARE

DIFFICULT

02 AI-Powered Underwater Marine Debris & Anomaly Detection via Side-Scan Sonar

SIH26057 · Ministry of Earth Sciences · Sustainability

Detects ghost fishing nets and debris in acoustic sonar imagery, where debris blends into natural seafloor features.

SOFTWARE

DIFFICULT

03 Real-Time Detection & Prevention of Voice Cloning Impersonation Attacks

SIH26104 · AICTE · Cybersecurity

AI voice cloning now lets scammers impersonate executives in real time; old caller-ID and callback checks no longer catch it.

SOFTWARE

DIFFICULT

04 Autonomous AI-Powered Search-and-Rescue Drone

SIH26177 · Qualcomm Inc · Robotics & Drones

An on-device AI drone that navigates disaster zones and detects survivors/hazards in real time with little to no connectivity.

HARDWARE

DIFFICULT

05 Adaptive Path Planning & Collision Avoidance on Unstructured Indian Roads

SIH26037 · MathWorks · Robotics & Drones

Standard autonomous-driving planners assume lane markings and predictable traffic — Indian roads offer neither.

SOFTWARE

DIFFICULT

Why these are hard: each needs either real hardware engineering under extreme conditions, live perception on unusual data (sonar, cloned audio), or full-stack robotics autonomy — genuinely research-adjacent work that a 36-hour team can only partially prototype.

DIFFICULT · DEEP DIVE

Problem → Why Hard → Approach → Tech Stack → MVP

What it actually takes to demo a credible slice of each of the 5 hardest picks in 36–120 hours.

01 Anti-Drone High-Altitude Optimization SIH26050 · DRDO PROBLEM Cold, thin air, wind and dust degrade the mechanical, RF and optical parts of anti-drone systems at altitude. WHY HARD Real defense hardware engineering across thermal, mechanical and RF domains — not a typical software build. APPROACH Simulate subsystem degradation under altitude stress and propose a hardened design fix. TECH STACK MATLAB/Simulink thermal-mechanical modeling, RF propagation simulation. MVP A simulation showing one subsystem's predicted degradation at altitude, with a fix proposal.	02 Underwater Marine Debris Detection SIH26057 · Ministry of Earth Sciences PROBLEM Ghost nets and debris litter the seafloor; manually scanning sonar logs for them is slow and error-prone. WHY HARD Computer vision on noisy acoustic imagery, where debris blends into natural rock/sand patterns. APPROACH Fine-tune a CV model on public side-scan sonar datasets to flag debris vs seafloor features. TECH STACK PyTorch/TensorFlow, sonar image preprocessing, transfer learning, GIS overlay. MVP A model flagging candidate debris regions on sample sonar image tiles.
03 Voice Cloning Attack Detection SIH26104 · AICTE PROBLEM AI-cloned voices now impersonate executives in real time to authorize fraud; old caller checks fail. WHY HARD Needs live, streaming deepfake-audio detection fast enough to flag a call mid-conversation. APPROACH Extract spectral/prosody artifacts synthetic voices leave, classify streaming audio chunks live. TECH STACK librosa feature extraction, lightweight CNN/transformer classifier, streaming pipeline. MVP A classifier flagging real vs AI-generated clips from a public benchmark in near real time.	04 Autonomous Search-and-Rescue Drone SIH26177 · Qualcomm Inc PROBLEM Responders need fast aerial intelligence in the first hours after a disaster, in damaged, low-connectivity terrain. WHY HARD Combines on-device AI, real-time flight autonomy and person/hazard detection — full robotics stack. APPROACH Onboard detection model + autonomous waypoint navigation + a live hazard overlay to ground. TECH STACK Edge AI board, YOLO-class detector, PX4/ArduPilot flight controller SITL simulation. MVP A simulated flight that autonomously detects a "person" and logs its GPS location.
05 Autonomous Vehicle Path Planning on Indian Roads SIH26037 · MathWorks PROBLEM Indian roads mix cars, two-wheelers, pedestrians and animals with unmarked lanes — the chaos that breaks path-planning algorithms designed for structured, predictable Western roads. WHY HARD Requires planning and collision-avoidance that works without lane markings or predictable motion assumptions — a genuinely open robotics problem. APPROACH Build a simulated unstructured-road environment, then design a planner reacting to unpredictable, mixed-speed obstacles in real time. TECH STACK MATLAB/Simulink Automated Driving Toolbox, ROS2 with simulated LIDAR/camera, sampling-based planners (RRT*). MVP A simulated vehicle safely navigating a small unstructured obstacle course.	

SIDE BY SIDE

All 10 picks at a glance

PS NO.	TITLE	ORGANIZATION	CATEGORY	LEVEL
SIH26044	Academia–Industry Collaboration Portal	Ministry of Ayush	Software	EASY
SIH26034	Packaged Commodities Compliance Checker	Consumer Affairs	Software	EASY
SIH26036	Instrument Verification System	Consumer Affairs	Software	EASY
SIH26063	Polar Science Outreach Portal	MoES (NCPOR)	Software	EASY
SIH26075	CAPACITY CONNECT LMS Portal	MoES	Software	EASY
SIH26050	Anti-Drone High-Altitude Optimization	DRDO	Hardware	DIFFICULT
SIH26057	Underwater Marine Debris Detection	MoES	Software	DIFFICULT
SIH26104	Voice Cloning Attack Detection	AICTE	Software	DIFFICULT
SIH26177	Autonomous Search-and-Rescue Drone	Qualcomm Inc	Hardware	DIFFICULT
SIH26037	Autonomous Vehicle Path Planning	MathWorks	Software	DIFFICULT

BEFORE YOU COMMIT

How to use this list

- First SIH team or under 48 hours of build time? Start from the Easy 5 — you can still add a smart feature (OCR, matching algorithm) to stand out.
- Team has shipped ML or embedded systems before? The Difficult 5 offer a far less crowded field and a much stronger "wow factor" on stage.
- Difficulty here is a heuristic, not official — always re-read the full PS on the SIH portal before finalising your idea.
- Whichever level you pick, scope your MVP down to the one feature above and get it fully working before adding anything else.

Match the PS to your team's **real** skill level — not your ambition.

Save this list, cross-check each PS Number on the official SIH 2026 portal, and lock your idea before the internal hackathon deadline.